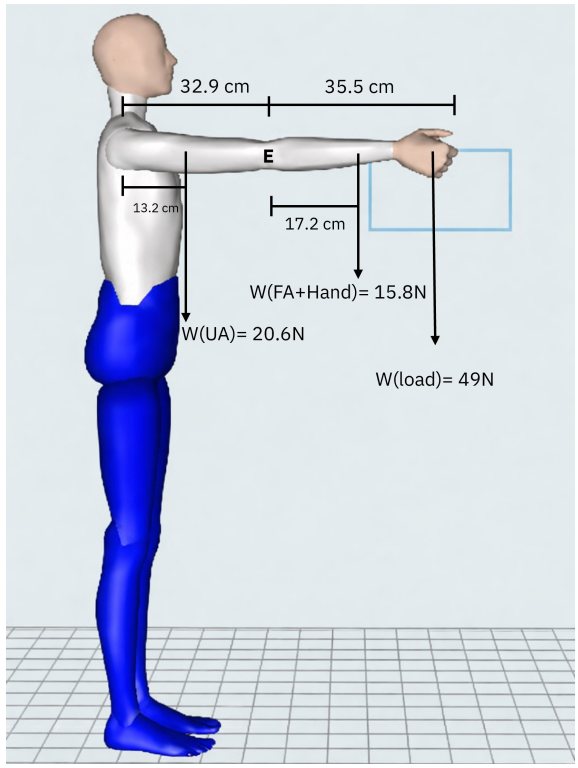
EIN3241 – Human Factors & Ergonomics

In class, we solved this example using the *segment-to-segment* approach through the elbow joint. Using that method, we determined that the reaction force and reaction moment at the shoulder were:

$$F_{\text{shoulder}} = 85.4 \text{ N}; M_{\text{shoulder}} = 44.15 \text{ Nm}$$



For this question, please redo the analysis using the **one-segment approach** to verify these results.

$$\sum F_y = 0$$

$$F_{\text{shoulder}} - W_{\text{UA}} - W_{\text{FA+H}} - W_{\text{load}} = 0$$

$$F_{\text{shoulder}} = 20.6 + 15.8 + 49 = 85.4N.$$

$$\boxed{F_{\text{shoulder}} \approx 85.4N \text{ (upward)}}$$

$$\sum M_{\text{shoulder}} = 0$$

All weights create a clockwise (negative) moment about the shoulder, so the shoulder reaction moment must balance them:

$$M_{\text{shoulder}} - \left(W_{\text{UA}} d_{\text{UA}} + W_{\text{FA+H}} d_{\text{FA+H}} + W_{\text{load}} d_{\text{load}}\right) = 0$$

$$M_{\text{shoulder}} = (20.6)(0.132) + (15.8)(0.501) + (49)(0.684)$$

$$M_{\text{shoulder}} = 2.7192 + 7.9158 + 33.516 = 44.151N.m$$

$$\boxed{M_{\text{shoulder}} \approx 44.15Nm \text{ (CCW)}}$$